

MTE 204 – Numerical Methods

Tutorial #3 Solutions

Question 1:

A manufacturing plant produces three different devices. Three materials (metal, plastic, and rubber) are required to produce each device. The amount of material needed for each device is detailed in the table below

Device	Metal (g/unit)	Plastic (g/unit)	Rubber (g/unit)
Device #1	15	0.25	1
Device #2	17	0.33	1.2
Device #3	19	0.42	1.6

The plant is consuming 2.12 kg of metal, 0.0434 kg of plastic, and 0.164 kg of rubber daily. Solve for the number of devices produced daily. Use gaussian elimination with partial pivoting.

Solution:

We begin by setting up our system of equations. In this case, we want to determine how many devices can be made for each source material daily. Note that material per device is given in grams, while total material usage is given in kilograms. The equations for metal, plastic, and rubber, respectively, are given below in units of grams.

$$\begin{aligned}15x_1 + 17x_2 + 19x_3 &= 2120 \\0.25x_1 + 0.33x_2 + 0.42x_3 &= 43.4 \\1x_1 + 1.2x_2 + 1.6x_3 &= 164\end{aligned}$$

Placing this system of equation in matrix form yields

$$\begin{bmatrix} 15 & 17 & 19 \\ 0.25 & 0.33 & 0.42 \\ 1 & 1.2 & 1.6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2120 \\ 43.4 \\ 164 \end{bmatrix}$$

Elimination – Column 1

Before we begin gaussian elimination to set all coefficients in the first column, except the pivot element, to zero, we must perform partial pivoting. Partial pivoting swaps two (and only two) rows to ensure the coefficient in the pivot element is the largest.

For the first column, the pivot element is a_{11} . The coefficient in the pivot element, 15, is already the largest of the three rows. Therefore, no pivoting is performed. Partial pivoting only swaps two rows; rows are not sorted.

Row 2:

$$\text{row 2} - \frac{a_{21}}{a_{11}} \text{row 1}$$

$$a_{2j} = \left[0.25 - \frac{0.25}{15} 15 \quad 0.33 - \frac{0.25}{15} 17 \quad 0.42 - \frac{0.25}{15} 19 \right]$$

$$a_{2j} = [0 \quad 0.04667 \quad 0.1033]$$

$$b_2 = 43.4 - \frac{0.25}{15} 2120 = 8.0667$$

Row 3

$$\text{row 3} - \frac{a_{31}}{a_{11}} \text{row 1}$$

$$a_{3j} = \left[1 - \frac{1}{15} 15 \quad 1.2 - \frac{1}{15} 17 \quad 1.6 - \frac{1}{15} 19 \right]$$

$$a_{3j} = [0 \quad 0.06667 \quad 0.3333]$$

$$b_3 = 164 - \frac{1}{15} 2120 = 22.6667$$

After the first iteration of elimination, we have the following matrix

$$\begin{bmatrix} 15 & 17 & 19 \\ 0 & 0.04667 & 0.1033 \\ 0 & 0.06667 & 0.3333 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2120 \\ 8.0667 \\ 22.6667 \end{bmatrix}$$

Elimination – Column 2

For the elimination of coefficients in column 3, the pivot element will be a_{22} . However, the coefficient in the pivot element, 0.04667, is not presently the largest. As such, we swap the second and third row to maximize the pivot element, yielding

$$\begin{bmatrix} 15 & 17 & 19 \\ 0 & 0.06667 & 0.3333 \\ 0 & 0.04667 & 0.1033 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2120 \\ 22.6667 \\ 8.0667 \end{bmatrix}$$

Remember, gaussian elimination seeks to **create an upper triangular matrix** and only eliminates coefficients below the current pivot element. Elements above are not affected.

Row 3

$$\text{row 3} - \frac{a_{32}}{a_{22}} \text{row 2}$$

$$a_{3j} = \left[0 \quad 0.04667 - \frac{0.04667}{0.06667} 0.06667 \quad 0.1033 - \frac{0.04667}{0.06667} 0.3333 \right]$$

$$a_{3j} = [0 \quad 0 \quad -0.13]$$

$$b_3 = 8.0667 - \frac{0.04667}{0.06667} 22.6667 = -7.8$$

After gaussian elimination, we have the following matrix.

$$\begin{bmatrix} 15 & 17 & 19 \\ 0 & 0.06667 & 0.3333 \\ 0 & 0 & -0.13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2120 \\ 22.6667 \\ -7.8 \end{bmatrix}$$

We can now perform back substitution

$$\begin{aligned} x_3 &= \frac{-7.8}{-0.13} = 60 \\ x_2 &= \frac{22.6667 - 0.3333(60)}{0.06667} = 40 \\ x_1 &= \frac{2120 - 19(60) - 17(40)}{15} = 20 \end{aligned}$$

Question 2:

Four springs of differing stiffness are placed in series and a **20 N force** is applied to the springs. The following equations are the force balances between each spring, with displacement defined relative to the fixed end of the springs.

$$\begin{aligned}k_2(x_2 - x_1) &= k_1 x_1 \\k_3(x_3 - x_2) &= k_2(x_2 - x_1) \\k_4(x_4 - x_3) &= k_3(x_3 - x_2) \\F &= k_4(x_4 - x_3)\end{aligned}$$

Let $k_1 = 15 \text{ N/cm}$, $k_2 = 5 \text{ N/cm}$, $k_3 = 8 \text{ N/cm}$, and $k_4 = 23 \text{ N/cm}$.

- Express the equations matrix form.
- Demonstrate that using the Gauss-Seidel method will result in convergence.
- Solve the problem iteratively using the Gauss-Seidel method.** Show the system of equations used to perform a Gauss-Seidel iteration. Carry out two iterations of the Gauss-Seidel method, showing you work for the first iteration. **Use 10 cm as an initial value for all displacements.**

Solution:

Part a:

Let begin by rearranging all equations to make placing them in matrix form simpler.

$$\begin{aligned}(k_1 + k_2)x_1 - k_2x_2 &= 0 \\-k_2x_1 + (k_2 + k_3)x_2 - k_3x_3 &= 0 \\-k_3x_2 + (k_3 + k_4)x_3 - k_4x_4 &= 0 \\-k_4x_3 + k_4x_4 &= F\end{aligned}$$

Placing these equations in matrix form yields

$$\begin{bmatrix} k_1 + k_2 & -k_2 & 0 & 0 \\ -k_2 & k_2 + k_3 & -k_3 & 0 \\ 0 & -k_3 & k_3 + k_4 & -k_4 \\ 0 & 0 & -k_3 & k_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ F \end{bmatrix}$$

Inserting numerical values yields

$$\begin{bmatrix} 20 & -5 & 0 & 0 \\ -5 & 13 & -8 & 0 \\ 0 & -8 & 31 & -23 \\ 0 & 0 & -23 & 23 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 20 \end{bmatrix}$$

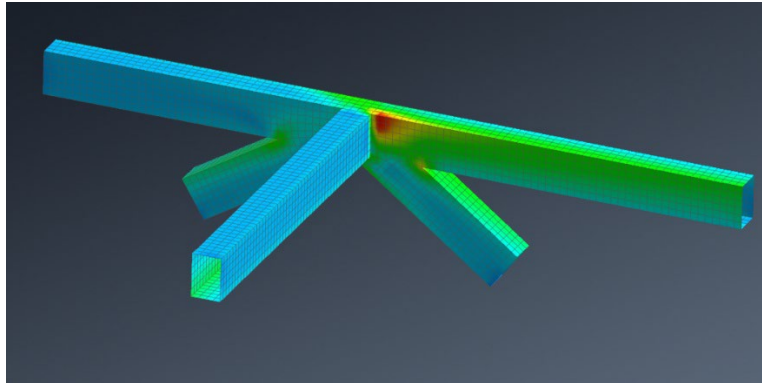
For trivia, this is known as a banded matrix, as it symmetric about the diagonal of the matrix.

It satisfies:

$$a_{ij} = a_{ji}$$

These type of matrix are actually very common in many situations (Like finite element analysis and heat transfer, and structural problems) and a number of **specialized algorithms** have been developed to solve

these types of problems quickly and efficiently. However, we will limit our focus to the Gauss-Seidel method.



In finite element analysis, the structure is divided into many small elements. Each node is usually connected only to neighbouring nodes. Therefore, the coefficient matrix often has many zeros, and the non-zero entries are concentrated near the diagonal.



In heat transfer or heat-treatment simulations, the temperature at one node is mainly affected by neighbouring nodes. After discretization, this local interaction leads to a banded coefficient matrix.

Part b:

Prior to carrying out Gauss-Seidel iteration, **we first need to demonstrate that this system of equations will converge using the Gauss-Seidel method.** This means the matrix must meet the condition below. In plain English, convergence is demonstrated if the element along the diagonal for a given row is larger than the sum of the absolute value of all other elements in the same row.

$$|a_{ii}| > \sum_{j=1, j \neq i}^n |a_{ij}|$$

Checking the convergence criteria yields

$$\begin{bmatrix} |a_{11}| \\ |a_{22}| \\ |a_{33}| \\ |a_{44}| \end{bmatrix} > \begin{bmatrix} |a_{12}| \\ |a_{21}| + |a_{23}| \\ |a_{32}| + |a_{34}| \\ |a_{43}| \end{bmatrix}$$

$$\begin{bmatrix} 20 \\ 13 \\ 31 \\ 23 \end{bmatrix} > \begin{bmatrix} 5 \\ 13 \\ 31 \\ 23 \end{bmatrix}$$

You will note that convergence is not strictly met for all equations. While this does not guarantee the system will converge, it also doesn't mean the system will not converge. Unfortunately, this is a somewhat contradictory case. **For the sake of this problem, let us continue under the assumption this system will converge.**

Part c:

Lets begin by setting up the equations need for **Gauss-Seidel iteration**. As a reminder, the general form of the equation used for this method is as follows

$$x_i = \left(\frac{1}{a_{ii}}\right) \left(b_i - \sum_{j=1, j \neq i}^n a_{ij} x_j\right)$$

The diagonal coefficient in each row should be larger than the sum of the other coefficients in that row.

$$\begin{bmatrix} 20 & -5 & 0 & 0 \\ -5 & 13 & -8 & 0 \\ 0 & -8 & 31 & -23 \\ 0 & 0 & -23 & 23 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 20 \end{bmatrix}$$

The four equations are (after substituting real values)

$$x_1 = \left(\frac{1}{20}\right) (0 - (-5)x_2) = \frac{5}{20} x_2$$

$$x_2 = \left(\frac{1}{13}\right) (0 - (-5)x_1 - (-8)x_3) = \frac{1}{13} (5x_1 + 8x_3)$$

$$x_3 = \left(\frac{1}{31}\right) (0 - (-8)x_2 - (-23)x_4) = \frac{1}{31} (8x_2 + 23x_4)$$

$$x_4 = \left(\frac{1}{23}\right) (20 - (-23x_3)) = \frac{1}{23} (20 + 23x_3)$$

Iteration 1

Now we can begin our first iteration. Remember, we are initializing all displacements to equal 10 cm.

For ease of writing, equations are organized in matrix form. Remember, we solve each equation incrementally and can use the newly calculated value of in subsequent equations. This is why the second equation below uses a value of 2.5 for x_1

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \frac{5}{20}(10) \\ \frac{1}{13}(5(2.5) + 8(10)) \\ \frac{1}{31}(8(7.115) + 23(10)) \\ \frac{1}{23}(20 + 23(9.2556)) \end{bmatrix} = \begin{bmatrix} 2.5 \\ 7.1154 \\ 9.2556 \\ 10.152 \end{bmatrix}$$

Iteration 2:

While the question only asks that one iteration be shown, we show the second iteration here in the hopes it will help to demonstrate how variables from earlier equations in an iteration can be used in the same iteration.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \frac{5}{20}(7.1154) \\ \frac{1}{13}(5(1.7789) + 8(9.2556)) \\ \frac{1}{31}(8(6.3799) + 23(10.1252)) \\ \frac{1}{23}(20 + 23(9.1587)) \end{bmatrix} = \begin{bmatrix} 1.7789 \\ 6.3799 \\ 9.1587 \\ 10.0283 \end{bmatrix}$$

The table below provides the first four Gauss-Seidel iterations for this system of equations, along with the relative absolute error for each variable.

$$|\epsilon_{a,i}| = \left| \frac{(x_{new,i} - x_{old,i})}{x_{new,i}} \right| \times 100\%$$

Iter.	x_1 (cm)	ϵ_{x_1} (%)	x_2 (cm)	ϵ_{x_2} (%)	x_3 (cm)	ϵ_{x_3} (%)	x_4 (cm)	ϵ_{x_4} (%)
0	10	-	10	-	10	-	10	-
1	2.5	300	7.1154	40.5	9.2556	8.04	10.1251	1.23
2	1.7788	40.5	6.3799	11.5	9.1586	1.06	10.0282	0.97
3	1.5920	11.5	6.2495	2.09	9.0531	1.17	9.9226	1.06
4	1.5624	2.09	6.1720	1.26	8.9547	1.10	9.8243	1.00

While a fixed number of iterations was carried out here, the inclusion of error calculations in the above table raises the question what the actual error of the iteration is. Or, phrased somewhat differently, if we were to allow the Gauss-Seidel method to converge automatically, what error metric should we use to gauge convergence? The textbook recommends taking the sum of all errors to use a termination criterion.

For reference, the actual solution to this system of equations is

$$x = \begin{bmatrix} 1.3333 \\ 5.3333 \\ 7.8333 \\ 8.7029 \end{bmatrix} \text{ cm}$$

Using the naïve Gauss-Seidel method (no relaxation, no pivoting), it takes approximately 66 iterations for the Gauss-Seidel method to converge to within two decimal places of this answer for all variables. This is very slow convergence behaviour. The use of relaxation (discussed in lecture) can help to improve convergence behaviour.

The solution does not apply relaxation here because the question asks for the naïve Gauss-Seidel method. The last sentence is just a note. Relaxation means modifying the Gauss-Seidel update by mixing the newly computed value with the previous value. If the relaxation factor is greater than 1, it may accelerate convergence. If it is less than 1, it may improve stability. In this example, the naïve method needs about 66 iterations, so relaxation could potentially reduce the number of iterations.